Shape-based Semantic Interoperability for Oncology Data Integration

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Abstract

Multiple isolated biomedical ontologies and clinical data standards coexist in the oncology domain, enabling rich semantic descriptions, while also introducing significant interoperability challenges. The lack of explicit and verifiable semantic constraints often leads to inconsistencies that hinder data reuse in clinical and research settings. This poster presents a shapes-based semantic interoperability approach that integrates heterogeneous oncology data sources using RDF and OWL, while enforcing structural and semantic consistency through SHACL and ShEx. The proposal is aligned with HL7 FHIR to ensure compatibility with clinical information systems and is demonstrated through a breast cancer case study, showing how data shapes support validation, interoperability, and reuse.

Keywords

Semantic Interoperability, Oncology Data, RDF Data Shapes, SHACL, ShEx, HL7 FHIR

1. Introduction

Oncology data are highly heterogeneous, covering EHRs, genomics, imaging, and research datasets [1]. Independent biomedical ontologies introduce semantic fragmentation and misalignment, while the absence of explicit validation rules limits interoperability and reuse.

Semantic Web standards (RDF, OWL) enable machine-interpretable knowledge, while SHACL and ShEx provide constraint languages for RDF validation [2]. HL7 FHIR acts as a bridge between semantic models and clinical systems [3].

This work presents a framework that integrates multiple ontologies, including NCIt, BCO, OntoBreast, and MAMO-SCR-Onto into a unified RDF/OWL knowledge graph for semantic representation [4]. Validation is performed using SHACL and ShEx shapes, executed with the rudof tool [5], while clinical alignment and interoperability are achieved through HL7 FHIR.

2. Case Study

This work focuses on the breast cancer domain, integrating data sources from SNOMED CT, NCIt, MeSH, BCO, and OntoBreast. Multiple breast cancer ontologies were integrated into a unified semantic ecosystem. The outcome is a validated RDF graph using ShEx and SHACL, aligned with HL7 FHIR resources.

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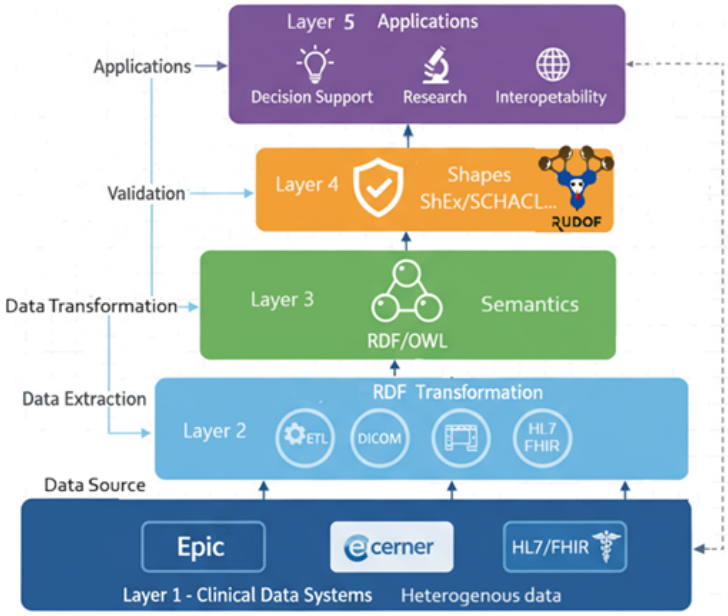
3. Architecture



The architecture follows a five-layer pipeline for semantic interoperability in cancer, ranging from heterogeneous clinical data sources to RDF transformation, semantic enrichment with OWL, validation using SHACL and ShEx, and final consumption by clinical and research applications.

4. Results and Conclusion

This approach contributes to improving semantic consistency in breast cancer datasets by enabling early detection of structural and semantic inconsistencies, while decoupling clinical systems from semantic validation logic. This work demonstrates that shape-based validation is an effective mechanism for achieving semantic interoperability in oncology. By combining RDF, OWL, SHACL, and ShEx with HL7 FHIR, the proposed approach supports data quality, reuse, and interoperability between clinical practice and biomedical research.

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